

STATEMENT OF WORK (SOW) & PROJECT PLAN

E-Commerce Ltd – AWS CLOUD & DEVOPS IMPLEMENTATION

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REVISION HISTORY

Version	Date	Prepared By	Status	Description
1.0	24-Apr-2026	Cloud Architecture Team	Draft	Initial version
1.1	24-Apr-2026	DevOps Practice Lead	Reviewed	Finalized version

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1. ABOUT DevOps Team

DevOps Team is a global leader in next-generation digital services and consulting, enabling enterprises to accelerate their cloud transformation journeys through scalable, secure, and resilient solutions.



DevOps Team provides end-to-end expertise across:

- Cloud Strategy & Migration
- DevOps & Automation
- Platform Engineering
- Application Modernization
- Managed Cloud Services

1.1 Value Proposition

- AWS Advanced Consulting Partner
 - Global delivery model with 24x7 support
 - Proven expertise in large-scale cloud transformations
 - Strong DevSecOps and automation capabilities
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2. PROJECT OVERVIEW

2.1 Executive Summary

The client intends to deploy a **production-grade Java-based web application** on AWS leveraging a **secure, scalable, and highly available 3-tier architecture**.

The proposed solution is designed to:

- Ensure **high availability across multiple Availability Zones**
- Enable **dynamic scalability using Auto Scaling Groups**
- Enforce **security best practices across all layers**
- Integrate **CI/CD pipelines for automated deployments**

 As illustrated in the architecture diagrams (*Page 1–2*), the solution adopts a **multi-tier architecture with clear separation of presentation, application, and data layers**, ensuring modular scalability and fault isolation

2.2 Scope of Work

DevOps Team will design and implement a cloud-native infrastructure on AWS for hosting the application, including:

- Network architecture design (VPC, subnets, routing)

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- Compute layer provisioning (EC2 with Auto Scaling)
 - Load balancing and traffic distribution
 - Database deployment and configuration (Amazon RDS)
 - CI/CD pipeline integration
 - Monitoring, logging, and alerting setup
 - Security hardening and compliance alignment
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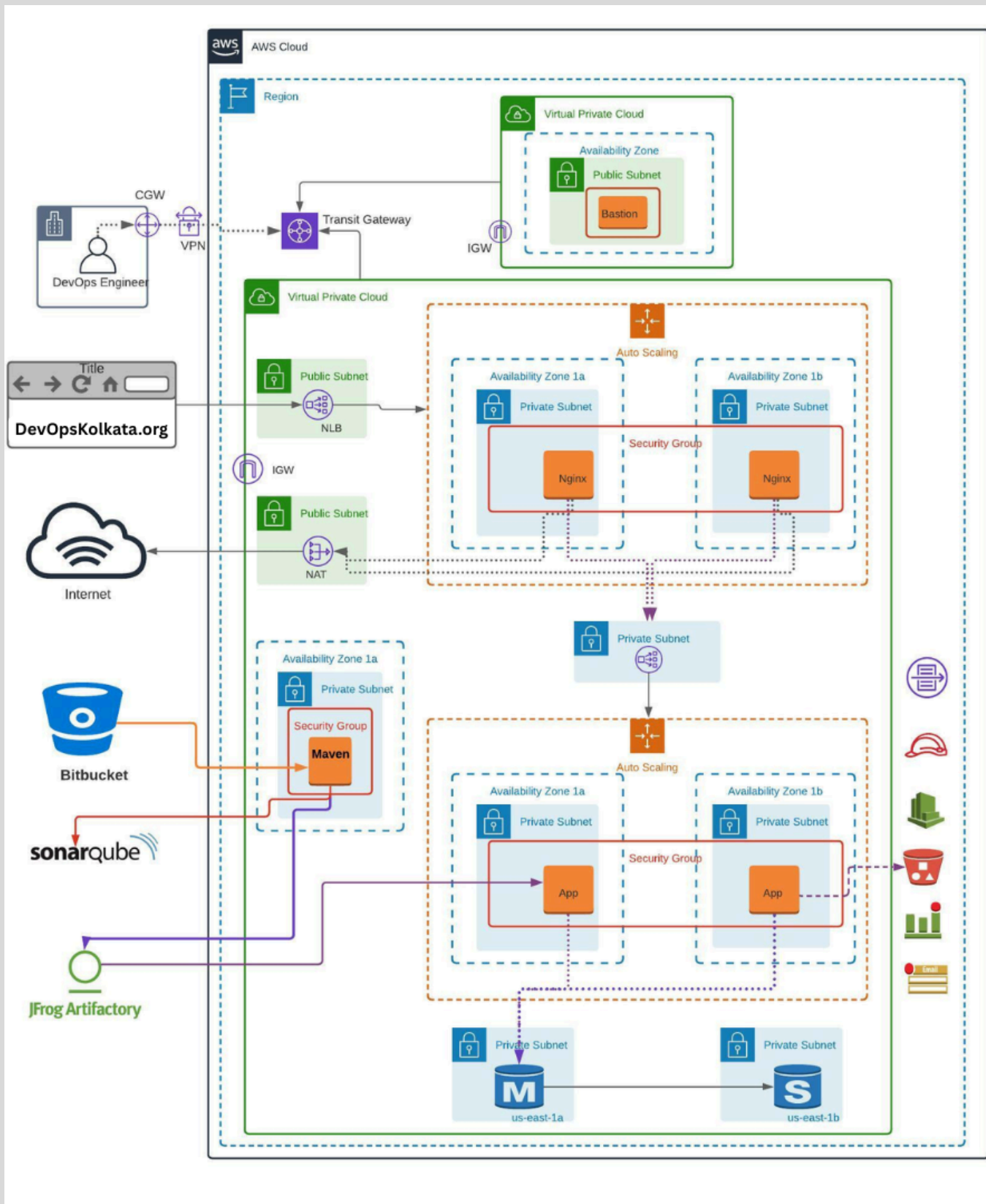
2.3 Customer Requirements

- Deploy Java application using a **3-tier architecture**
- Implement **high availability and fault tolerance**
- Enable **auto scaling based on workload demand**
- Integrate **DevOps tools for CI/CD**
- Ensure **secure and compliant infrastructure**

 The project leverages tools such as Nginx, Apache Tomcat, Amazon RDS, SonarQube, and artifact repositories for a complete DevOps lifecycle (*Page 2*)

3. SOLUTION ARCHITECTURE & IMPLEMENTATION

3.1 Target Architecture on AWS





The proposed architecture consists of the following layers:

Presentation Layer

- Nginx Web Servers deployed in Auto Scaling Groups
- Internet-facing Application Load Balancer
- CloudFront (optional) for content delivery

Application Layer

- Apache Tomcat servers hosted on EC2 instances
- Internal Load Balancer for service communication
- Stateless application design for scalability

Data Layer

- Amazon RDS (MySQL) in Multi-AZ configuration
- Automated backups and failover support
- Optional caching layer using Amazon ElastiCache

DevOps & Integration Layer

- Source Code Management (Git-based repository)
- Code Quality Analysis (SonarQube)
- Artifact Management (JFrog Artifactory)

 The architecture diagram (*Page 2*) highlights segregation of public and private subnets, secure communication between layers, and integration with CI/CD tools

3.2 Implementation Approach

The implementation will be executed in structured phases:

Phase 1: Network & Foundation Setup

- Provision Virtual Private Cloud (VPC)
- Configure public and private subnets across multiple AZs
- Attach Internet Gateway and configure NAT Gateway

- Establish route tables and network segmentation

 Refer to infrastructure setup steps (*Page 3–4*)

Phase 2: Security Configuration

- Define Security Groups and Network ACLs
 - Configure IAM roles and policies with least privilege access
 - Implement secure access mechanisms (SSH via bastion or VPN)
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Phase 3: Database Deployment

- Deploy Amazon RDS MySQL instance (Multi-AZ)
 - Configure database parameters and storage
 - Enable automated backups and monitoring
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Phase 4: Application Deployment

- Install and configure:
 - Apache Tomcat (backend services)
 - Nginx (frontend proxy)
 - Deploy application artifacts
 - Configure reverse proxy routing
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Phase 5: Load Balancing & Auto Scaling

- Configure Application Load Balancer (ALB)
- Create target groups and health checks
- Implement Auto Scaling Groups with dynamic scaling policies

 Refer to load balancing and scaling setup (*Page 6*)

Phase 6: CI/CD Integration

- Integrate source repository with pipeline

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- Configure SonarQube for code quality checks
 - Setup artifact repository for build storage
 - Enable automated deployment pipelines
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Phase 7: Monitoring & Observability

- Configure Amazon CloudWatch for metrics and logs
- Set up alarms for system health and performance
- Implement centralized logging strategy

 Monitoring setup referenced in *(Page 6)*

Phase 8: Security & Compliance

- Enable encryption at rest and in transit
- Implement AWS WAF and Shield (if required)
- Enable VPC Flow Logs and audit logging
- Apply security best practices across all layers

 Security recommendations outlined in *(Page 7)*

3.3 Out of Scope

- Application code development or modification
 - Data migration from legacy systems
 - Performance benchmarking beyond infrastructure layer
 - Third-party integrations not explicitly defined
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3.4 Assumptions

- Customer will provide required AWS account access
 - Application is production-ready and tested
 - Required credentials for CI/CD tools will be shared
 - Stakeholders will be available for timely approvals
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3.5 RACI Matrix

Activity	Responsible	Accountable	Consulted	Informed
Architecture Design	DevOps Team	DevOps Team	Customer	Customer
Implementation	DevOps Team	DevOps Team	AWS	Customer
Validation	DevOps Team	Customer	DevOps Team	AWS
Final Sign-off	Customer	Customer	DevOps Team	AWS

4. GOVERNANCE, RESOURCES & COST ESTIMATES

4.1 Resource Model

Role	Responsibility
Cloud Architect	Solution design and governance
DevOps Engineer	Implementation & automation
Cloud Engineer	Infrastructure provisioning
Project Manager	Delivery coordination

4.2 Service Levels (Indicative SLA)

Priority	Description	Response Time	Resolution Time
P1	Critical outage	15 mins	4 hours

P2	Major issue	30 mins	8 hours
P3	Standard request	2 hours	2 business days

4.3 Estimated Infrastructure Cost

- Monthly AWS Cost: **USD 150 – 300** (indicative)
- Managed Services Cost: **~30% of monthly cloud spend**

AWS Calculator Link:

Final cost may vary based on actual usage, scaling patterns, and data transfer.

5. TERMS AND CONDITIONS

5.1 Change Management

Any deviation from the agreed scope will be governed through a formal Change Request (CR) process, including impact analysis and approval.

5.2 Jurisdiction

This agreement shall be governed under the applicable laws of India.

5.3 Confidentiality

Both parties agree to maintain confidentiality of all project-related data and intellectual property.

5.4 Termination

Either party may terminate this agreement with a **30-day written notice**.

6. ACCEPTANCE OF PROPOSAL

By signing below, both parties agree to the scope, terms, and conditions outlined in this document.

Customer Representative	DevOps Team
Name: _____	Name: _____
Signature: _____	Signature: _____
Date: _____	Date: _____